HCIA-AI V4.0 — Python, Frameworks & Core Concepts Fill-in-the-Blank Guide

Authentic Exam Practice • 70 High-Yield Fill-in-the-Blank Questions (16 Points Each)

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OFFICIAL EXAM FILL-IN-THE-BLANK QUESTION FORMAT & CORE CONVENTIONS

1. Exam Format (16 Points): Questions present technical statements with ____. Write the exact method, class, or constant on the answer line.	**2. Framework Naming:** PyTorch subclasses torch.nn.Module & uses forward(); MindSpore subclasses mindspore.nn.Cell & uses construct().
3. Data Preprocessing: Always run fit_transform(X_train) on training data, and strictly run transform(X_test) on test data to avoid data leakage.	**4. Core Constants:** Sigmoid max derivative = **0.25** (at x=0); Softmax sum = **1**; ReLU for x<0 = **0**; Ascend 310 = **8W** edge, Ascend 910 = **310W** cloud.

Category 1: PyTorch Deep Learning Workflow & API (Questions 1 – 15)

Question 1

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, all custom neural network models and custom layers must inherit from the base class torch.nn.____.

Answer: _____

Question 2

FILL-IN-THE-BLANK QUESTION • 16 Points

In a custom PyTorch nn.Module class, the method that defines the forward computation pass of input data x is named def ____(self, x):.

Answer: _____

Question 3

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to prevent gradients from accumulating across consecutive training iterations, gradients must be reset before backward pass using optimizer.____().

Answer: _____

Question 4

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, after calculating the loss tensor, the autograd method invoked on the loss tensor to compute gradients of all parameters is loss.____().

Answer: _____

Page Answers: **Q1**: Module • **Q2**: forward • **Q3**: zero_grad • **Q4**: backward

Question 5

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the optimizer method called to update neural network weights and biases according to computed gradients is optimizer.____().

Answer: _____

Question 6

FILL-IN-THE-BLANK QUESTION • 16 Points

During PyTorch model evaluation and testing, the autograd engine is temporarily deactivated to conserve memory using the context manager with torch.____():.

Answer: _____

Question 7

FILL-IN-THE-BLANK QUESTION • 16 Points

To notify PyTorch layers like Dropout and BatchNorm that the network is entering inference mode, the method invoked on the model is model.____().

Answer: _____

Question 8

FILL-IN-THE-BLANK QUESTION • 16 Points

To switch a PyTorch model back into training mode so Dropout and BatchNorm operate normally, the method called is model.____().

Answer: _____

Question 9

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the utility class from torch.utils.data that wraps a Dataset to provide mini-batching, shuffling, and multi-process loading is ____.

Answer: _____

Question 10

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to enable automatic gradient computation on an input tensor during the forward pass, the tensor attribute must be set to ____=True.

Answer: _____

Page Answers: **Q5**: step • **Q6**: no_grad • **Q7**: eval • **Q8**: train • **Q9**: DataLoader • **Q10**: requires_grad

Question 11

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the universal method called on tensors or models to transfer them to a target computing device (such as 'cuda' or 'cpu') is .____(device).

Answer: _____

Question 12

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to extract the standard Python numerical value from a single-element loss tensor (preventing CUDA memory leaks), you invoke loss.____().

Answer: _____

Question 13

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the standard loss criterion that combines LogSoftmax and NLLLoss in a single step for multi-class classification is nn.____().

Answer: _____

Question 14

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the standard loss criterion used to compute the mean squared error for continuous numeric regression tasks is nn.____().

Answer: _____

Question 15

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the tensor method used to reshape or flatten a multidimensional tensor into a 2D matrix before entering fully connected layers is tensor.____(batch_size, -1).

Answer: _____

Category 2: Huawei MindSpore Framework, DaVinci Architecture & Ascend (Questions 16 – 30)

Question 16

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, all custom neural network models and custom layers must inherit from the base class mindspore.nn.____.

Answer: _____

Page Answers: **Q11**: to • **Q12**: item • **Q13**: CrossEntropyLoss • **Q14**: MSELoss • **Q15**: view • **Q16**: Cell

Question 17

FILL-IN-THE-BLANK QUESTION • 16 Points

Inside a custom MindSpore nn.Cell subclass, the computational execution graph for the forward pass must be defined in the def ____(self, x): method.

Answer: _____

Question 18

FILL-IN-THE-BLANK QUESTION • 16 Points

In MindSpore, the compilation mode that constructs a static computation graph for maximum execution efficiency on Ascend hardware is ms.____.

Answer: _____

Question 19

FILL-IN-THE-BLANK QUESTION • 16 Points

In MindSpore, the interactive execution mode that executes operators immediately line-by-line for easy debugging is ms.____.

Answer: _____

Question 20

FILL-IN-THE-BLANK QUESTION • 16 Points

In mindspore.set_context(), the parameter used to specify the execution hardware platform ('Ascend', 'GPU', or 'CPU') is named ____.

Answer: _____

Question 21

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, the neural network layer class that implements a fully connected feedforward linear transformation is nn.____.

Answer: _____

Question 22

FILL-IN-THE-BLANK QUESTION • 16 Points

The top-level MindSpore function used to serialize neural network parameters into a checkpoint file on disk is ms.____.

Answer: _____

Page Answers: **Q17**: construct • **Q18**: GRAPH_MODE • **Q19**: PYNATIVE_MODE • **Q20**: device_target • **Q21**: Dense • **Q22**: save_checkpoint

Question 23

FILL-IN-THE-BLANK QUESTION • 16 Points

The MindSpore function used to read a serialized .ckpt file from disk and parse it into a parameter dictionary is ms.____.

Answer: _____

Question 24

FILL-IN-THE-BLANK QUESTION • 16 Points

After loading a parameter dictionary in MindSpore, the function used to load those parameters directly into the live neural network Cell is ms.____.

Answer: _____

Question 25

FILL-IN-THE-BLANK QUESTION • 16 Points

In modern MindSpore, the function used to create a gradient computation callable that returns both forward loss and parameter gradients is ms.____.

Answer: _____

Question 26

FILL-IN-THE-BLANK QUESTION • 16 Points

When preprocessing image datasets in mindspore.dataset, the operation that applies transformation pipelines across dataset columns is dataset.____().

Answer: _____

Question 27

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's AI processor series built on the DaVinci architecture is named _____.

Answer: _____

Question 28

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's AI processor series specifically designed for low-power edge inference (consuming 8W) is named Ascend ____.

Answer: _____

Page Answers: **Q23**: load_checkpoint • **Q24**: load_param_into_net • **Q25**: value_and_grad • **Q26**: map • **Q27**: Ascend • **Q28**: 310

Question 29

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's flagship AI processor series designed for ultra-high-throughput cloud training (consuming 310W) is named Ascend ____.

Answer: _____

Question 30

FILL-IN-THE-BLANK QUESTION • 16 Points

The chip enablement software architecture layer that bridges Huawei Ascend hardware with AI frameworks is called CANN, which stands for Compute Architecture for ____ Networks.

Answer: _____

Category 3: Scikit-Learn Traditional Machine Learning (Questions 31 – 45)

Question 31

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, the standard function from sklearn.model_selection used to randomly partition data into training and testing sets is ____.

Answer: _____

Question 32

FILL-IN-THE-BLANK QUESTION • 16 Points

In the universal Scikit-Learn Estimator API, the core method called to train an algorithm on feature matrix X and target labels y is model.____(X, y).

Answer: _____

Question 33

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, after fitting an estimator, the method used to generate discrete predicted class labels for new unseen samples is model.____(X_test).

Answer: _____

Question 34

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn classification estimators (such as LogisticRegression), the method that returns calibrated class probability estimates is model.____(X_test).

Answer: _____

Page Answers: **Q29**: 910 • **Q30**: Neural • **Q31**: train_test_split • **Q32**: fit • **Q33**: predict • **Q34**: predict_proba

Question 35

FILL-IN-THE-BLANK QUESTION • 16 Points

When normalizing feature data using transformers, the method called strictly on the training set to compute statistics and scale features in one step is scaler.____(X_train).

Answer: _____

Question 36

FILL-IN-THE-BLANK QUESTION • 16 Points

To prevent data leakage when normalizing test features, the transformer method that applies previously learned training parameters without refitting is scaler.____(X_test).

Answer: _____

Question 37

FILL-IN-THE-BLANK QUESTION • 16 Points

The scikit-learn preprocessing class that standardizes features by removing the mean and scaling to unit variance (z-score) is ____.

Answer: _____

Question 38

FILL-IN-THE-BLANK QUESTION • 16 Points

The scikit-learn transformer that scales and shifts features into a bounded range, typically between 0 and 1, is ____.

Answer: _____

Question 39

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn hyperparameter tuning, the class that performs an exhaustive search over every combination in a specified parameter grid is ____.

Answer: _____

Question 40

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn hyperparameter tuning, the class that samples a fixed number of parameter candidates from specified statistical distributions is ____.

Answer: _____

Page Answers: **Q35**: fit_transform • **Q36**: transform • **Q37**: StandardScaler • **Q38**: MinMaxScaler • **Q39**: GridSearchCV • **Q40**: RandomizedSearchCV

Question 41

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, the utility used to sequentially assemble several preprocessing transformers and a final estimator into a single unified object is ____.

Answer: _____

Question 42

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn classification metrics, the function that computes the fraction of correct predictions over total predictions is ____.

Answer: _____

Question 43

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn classification evaluation, the function that computes a contingency table showing True Positives, False Positives, False Negatives, and True Negatives is ____.

Answer: _____

Question 44

FILL-IN-THE-BLANK QUESTION • 16 Points

The evaluation metric computed as the harmonic mean of precision and recall is the ____ score.

Answer: _____

Question 45

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn unsupervised learning, the standard algorithm from sklearn.cluster that partitions samples into K distinct clusters minimizing within-cluster variance is ____.

Answer: _____

Category 4: NumPy & Tensor Algebraic Foundations (Questions 46 – 55)

Question 46

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function or operator used to perform linear algebraic matrix multiplication between two 2D arrays A and B is np.____(A, B) or A @ B.

Answer: _____

Page Answers: **Q41**: Pipeline • **Q42**: accuracy_score • **Q43**: confusion_matrix • **Q44**: F1 • **Q45**: KMeans • **Q46**: dot

Question 47

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy and PyTorch, the attribute or method that reverses or permutes the axes of a matrix (swapping rows and columns) is matrix.____.

Answer: _____

Question 48

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the ndarray attribute that returns a tuple representing the sizes of each dimension of the array is array.____.

Answer: _____

Question 49

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the ndarray attribute that returns the number of array dimensions (rank / tensor order) is array.____.

Answer: _____

Question 50

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the ndarray attribute that describes the memory data type of the elements (such as float32, int64) is array.____.

Answer: _____

Question 51

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function that creates a new array of a given shape and type, filled entirely with zeros, is np.____(shape).

Answer: _____

Question 52

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function that creates a new array of a given shape filled entirely with ones is np.____(shape).

Answer: _____

Page Answers: **Q47**: T • **Q48**: shape • **Q49**: ndim • **Q50**: dtype • **Q51**: zeros • **Q52**: ones

Question 53

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function used to generate a 2D identity matrix with ones on the main diagonal and zeros elsewhere is np.____(N).

Answer: _____

Question 54

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function used to join a sequence of existing arrays along an existing axis is np.____((a1, a2), axis=0).

Answer: _____

Question 55

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function that returns the indices of the maximum values along a specified axis is np.____(array, axis=1).

Answer: _____

Category 5: Deep Learning Dimensions, Formulas & Constants (Questions 56 – 70)

Question 56

FILL-IN-THE-BLANK QUESTION • 16 Points

For the Sigmoid activation function σ(x) = 1 / (1 + e^-x), the maximum theoretical value of its derivative σ'(x) occurs at x = 0 and equals ____.

Answer: _____

Question 57

FILL-IN-THE-BLANK QUESTION • 16 Points

For a multi-class neural network with a Softmax output layer, the mathematical sum of all output probabilities across all classes strictly equals ____.

Answer: _____

Question 58

FILL-IN-THE-BLANK QUESTION • 16 Points

For the Rectified Linear Unit (ReLU) activation function f(x) = max(0, x), the mathematical output value for any negative input (x < 0) is ____.

Answer: _____

Page Answers: **Q53**: eye • **Q54**: concatenate • **Q55**: argmax • **Q56**: 0.25 • **Q57**: 1 • **Q58**: 0

Question 59

FILL-IN-THE-BLANK QUESTION • 16 Points

The deep learning activation function whose output values are symmetrically bounded in the range [-1, +1] is ____.

Answer: _____

Question 60

FILL-IN-THE-BLANK QUESTION • 16 Points

In deep feedforward networks, when activation functions like Sigmoid repeatedly squash derivatives, causing early layers to stop learning, this is called the ____ gradient problem.

Answer: _____

Question 61

FILL-IN-THE-BLANK QUESTION • 16 Points

When derivatives in deep networks multiply and grow uncontrollably large during backpropagation, causing numeric overflow (NaNs), this is called the ____ gradient problem.

Answer: _____

Question 62

FILL-IN-THE-BLANK QUESTION • 16 Points

The regularization technique that adds the sum of absolute values of weights (||w||_1) to the loss function, encouraging parameter sparsity, is ____ regularization.

Answer: _____

Question 63

FILL-IN-THE-BLANK QUESTION • 16 Points

The regularization technique that adds the sum of squared weights (||w||_2^2) to the loss function, commonly referred to as weight decay, is ____ regularization.

Answer: _____

Question 64

FILL-IN-THE-BLANK QUESTION • 16 Points

The deep learning regularization technique that randomly sets a fraction p of hidden neuron activations to zero during training is named ____.

Answer: _____

Page Answers: **Q59**: tanh • **Q60**: vanishing • **Q61**: exploding • **Q62**: L1 • **Q63**: L2 • **Q64**: Dropout

Question 65

FILL-IN-THE-BLANK QUESTION • 16 Points

In convolutional neural networks (CNNs), the step size (in pixels) by which the sliding kernel filter traverses the input feature map is called the ____.

Answer: _____

Question 66

FILL-IN-THE-BLANK QUESTION • 16 Points

In CNNs, adding border pixels (typically zeros) around the input matrix to prevent feature map shrinkage is called ____.

Answer: _____

Question 67

FILL-IN-THE-BLANK QUESTION • 16 Points

The CNN operation (such as Max or Average) that subsamples spatial dimensions while preserving depth and translational invariance is called ____.

Answer: _____

Question 68

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's computing hardware platform that includes AI accelerator modules, PCIe cards, edge computing stations, and AI training/inference servers is named ____.

Answer: _____

Question 69

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei Cloud, the one-stop, full-lifecycle AI development and management service covering data labeling, training, and API deployment is named ____.

Answer: _____

Question 70

FILL-IN-THE-BLANK QUESTION • 16 Points

The universal intermediate open representation format used to export PyTorch models for deployment across heterogeneous hardware (including Huawei Ascend via CANN) is ____.

Answer: _____

Page Answers: **Q65**: stride • **Q66**: padding • **Q67**: pooling • **Q68**: Atlas • **Q69**: ModelArts • **Q70**: ONNX

HCIA-AI V4.0 Master Quick-Review Answer Key: Questions 1 – 35

#	Verified Correct Answer	Technical Explanation, Key Formula & Official Exam Tip (Questions 1 – 35)
Q1	Module	[PyTorch Workflow] Fundamental building block in PyTorch. All neural network architectures inherit from torch.nn.Module.
Q2	forward	[PyTorch Workflow] Defines forward execution logic; PyTorch's __call__ automatically wraps forward and manages hooks.
Q3	zero_grad	[PyTorch Workflow] PyTorch autograd accumulates gradients by default; optimizer.zero_grad() flushes them to zero before backpropagation.
Q4	backward	[PyTorch Workflow] Executes backward computation graph, computing partial derivatives for all tensors with requires_grad=True.
Q5	step	[PyTorch Workflow] Updates parameters in-place using designated optimization algorithm (e.g. SGD, Adam, AdamW).
Q6	no_grad	[PyTorch Workflow] Disables gradient tracking history, accelerating forward execution and conserving GPU VRAM during validation.
Q7	eval	[PyTorch Workflow] Freezes BatchNorm running statistics and disables Dropout stochastic neuron deactivation for deterministic evaluation.
Q8	train	[PyTorch Workflow] Restores training behavior for layers with train/eval branching logic (Dropout, BatchNorm, LayerNorm).
Q9	DataLoader	[PyTorch Workflow] Yields minibatches from torch.utils.data.Dataset; supports multiprocessing, pin_memory, and customized collate functions.
Q10	requires_grad	[PyTorch Workflow] Flags tensor for autograd tracking, constructing backward computational graph nodes dynamically.
Q11	to	[PyTorch Workflow] Moves memory buffers across devices in-place or copies tensors (e.g., model.to('cuda')).
Q12	item	[PyTorch Workflow] Extracts raw Python float; prevents keeping computational graph history in GPU memory across epochs.
Q13	CrossEntropyLoss	[PyTorch Workflow] Takes unnormalized logits directly; combines LogSoftmax and Negative Log-Likelihood Loss with numerical stability.
Q14	MSELoss	[PyTorch Workflow] Computes squared L2 norm between target continuous values and predictions: (1/N) * sum((y - y_hat)^2).
Q15	view	[PyTorch Workflow] Returns new tensor with same data in new shape; requires memory contiguity (or use .reshape()).
Q16	Cell	[MindSpore & Ascend] Fundamental building block in MindSpore (analogous to PyTorch's nn.Module).
Q17	construct	[MindSpore & Ascend] MindSpore specifies construct (PyTorch specifies forward). Both implement forward propagation logic.
Q18	GRAPH_MODE	[MindSpore & Ascend] Static graph compilation optimizes operators on Ascend/GPU; PYNATIVE_MODE enables dynamic execution.
Q19	PYNATIVE_MODE	[MindSpore & Ascend] Dynamic eager execution mode in MindSpore, ideal for Python debugging and research prototyping.
Q20	device_target	[MindSpore & Ascend] Accepts 'Ascend', 'GPU', or 'CPU', routing operator execution to the designated hardware accelerator.
Q21	Dense	[MindSpore & Ascend] MindSpore designates fully connected layers as nn.Dense (PyTorch designates them as nn.Linear).
Q22	save_checkpoint	[MindSpore & Ascend] Writes Cell parameters to disk in Huawei's native checkpoint format (e.g. lenet.ckpt).
Q23	load_checkpoint	[MindSpore & Ascend] Reads the serialized checkpoint file and returns a dictionary mapping parameter names to tensors.
Q24	load_param_into_net	[MindSpore & Ascend] Matches parameter dictionary keys with target Cell layer weights and updates model weights in-place.
Q25	value_and_grad	[MindSpore & Ascend] Creates a callable returning (loss, grads), streamlining gradient updates in PyNative mode.
Q26	map	[MindSpore & Ascend] Executes transformation pipelines (e.g. Resize, ToTensor, Normalize) across specified columns.
Q27	Ascend	[MindSpore & Ascend] Official Huawei AI processor family powered by proprietary DaVinci 3D Cube computing architecture.
Q28	310	[MindSpore & Ascend] Ascend 310 is dedicated to Edge Inference (8W); Ascend 910 is dedicated to Cloud Training (310W).
Q29	910	[MindSpore & Ascend] Ascend 910 delivers half-precision (FP16) compute for deep neural network cloud training.
Q30	Neural	[MindSpore & Ascend] CANN (Compute Architecture for Neural Networks) provides AscendCL APIs and TBE operator acceleration.
Q31	train_test_split	[Scikit-Learn ML] Randomly partitions feature matrices X and labels y while preserving sample index alignment.
Q32	fit	[Scikit-Learn ML] All Scikit-Learn estimators implement .fit() to calculate and store learned internal parameters.
Q33	predict	[Scikit-Learn ML] Returns categorical class predictions. For continuous regression, returns continuous float predictions.
Q34	predict_proba	[Scikit-Learn ML] Returns array of shape (n_samples, n_classes) with normalized probability scores summing to 1.0.
Q35	fit_transform	[Scikit-Learn ML] Combines .fit() (computes mean and std) and .transform() (scales data). Strictly run on train only.

Page Answers: *Master Answer Key & Deep-Dive Technical Explanations*

HCIA-AI V4.0 Master Quick-Review Answer Key: Questions 36 – 70

#	Verified Correct Answer	Technical Explanation, Key Formula & Official Exam Tip (Questions 36 – 70)
Q36	transform	[Scikit-Learn ML] Applies learned training scaling parameters (mean/variance) to test samples without computing new statistics.
Q37	StandardScaler	[Scikit-Learn ML] Transforms features to zero mean and unit variance: z = (x - u) / s. Sensitive to extreme outliers.
Q38	MinMaxScaler	[Scikit-Learn ML] Linearly transforms features: (x - min) / (max - min). Scales features into strict interval [0, 1].
Q39	GridSearchCV	[Scikit-Learn ML] Evaluates every Cartesian combination in parameter grid; guarantees finding best combination within discrete space.
Q40	RandomizedSearchCV	[Scikit-Learn ML] Draws random samples from parameter distributions across fixed n_iter; faster for high-dimensional search.
Q41	Pipeline	[Scikit-Learn ML] Chains transformers and estimators; prevents data leakage during cross-validation by keeping transformations encapsulated.
Q42	accuracy_score	[Scikit-Learn ML] Accuracy = (TP + TN) / (TP + TN + FP + FN). Reliable only on balanced class distributions.
Q43	confusion_matrix	[Scikit-Learn ML] Evaluates classification performance across actual vs predicted labels; foundational for Precision, Recall, and F1.
Q44	F1	[Scikit-Learn ML] F1 = 2 * (Precision * Recall) / (Precision + Recall). Balances false positives and false negatives under class imbalance.
Q45	KMeans	[Scikit-Learn ML] Iterative partitioning clustering algorithm that minimizes Within-Cluster Sum of Squares (WCSS inertia).
Q46	dot	[NumPy & Algebra] Computes dot product / matrix multiplication; @ operator is syntactic sugar for np.matmul.
Q47	T	[NumPy & Algebra] Transposes 2D arrays; swaps axes 0 and 1 without copying underlying memory buffer.
Q48	shape	[NumPy & Algebra] Tuple representing array dimensions: e.g., (3, 224, 224) indicates channels, height, and width.
Q49	ndim	[NumPy & Algebra] Returns integer rank: scalar=0, vector=1, matrix=2, 3D tensor=3.
Q50	dtype	[NumPy & Algebra] Determines precision and byte storage of elements (e.g. np.float32, np.int64).
Q51	zeros	[NumPy & Algebra] Allocates contiguous memory filled with 0.0; fundamental for weight and bias initialization.
Q52	ones	[NumPy & Algebra] Allocates array populated entirely with 1.0; used for mask matrices and scale initialization.
Q53	eye	[NumPy & Algebra] Generates identity matrix I of shape (N, N); fundamental for linear algebraic transformations.
Q54	concatenate	[NumPy & Algebra] Joins arrays along specified axis; requires matching shapes along all non-concatenation dimensions.
Q55	argmax	[NumPy & Algebra] Returns integer index of peak value; used to convert Softmax probability vectors into predicted class IDs.
Q56	0.25	[Formulas & Constants] Since σ'(x) = σ(x)(1 - σ(x)), at x=0, σ(0)=0.5, so max derivative is 0.5 * 0.5 = 0.25. Causes vanishing gradients.
Q57	1	[Formulas & Constants] Softmax normalizes unconstrained exponential logits such that sum(P_i) = 1.0, satisfying probability distribution axioms.
Q58	0	[Formulas & Constants] ReLU clamps negative inputs to 0; prevents vanishing gradients for positive activations but can cause 'dying ReLU'.
Q59	tanh	[Formulas & Constants] Zero-centered hyperbolic tangent activation; derivative max is 1.0 (at x=0), but still suffers from saturation.
Q60	vanishing	[Formulas & Constants] Backpropagated gradient shrinks exponentially as depth increases; solved by ReLU and residual connections.
Q61	exploding	[Formulas & Constants] Solved by gradient clipping (clip_grad_norm_), weight regularization, and proper weight initialization (He/Xavier).
Q62	L1	[Formulas & Constants] L1 (Lasso) penalty forces non-essential feature weights to exactly zero, performing automated feature selection.
Q63	L2	[Formulas & Constants] L2 (Ridge / weight decay) shrinks weights smoothly towards zero without driving them strictly to zero.
Q64	Dropout	[Formulas & Constants] Prevents complex co-adaptation of neurons; during test mode (eval), weights are scaled by (1 - p) with no dropout.
Q65	stride	[Formulas & Constants] Controls downsampling; Stride=1 keeps resolution, Stride=2 halves feature map spatial height and width.
Q66	padding	[Formulas & Constants] 'Same' padding preserves input spatial dimensions: Pad = (Kernel - 1) / 2; 'Valid' padding adds zero borders.
Q67	pooling	[Formulas & Constants] Max Pooling extracts dominant spatial features; Average Pooling smooths features; both reduce parameter count.
Q68	Atlas	[Formulas & Constants] Atlas hardware infrastructure portfolio: Atlas 200/300/500/800/900 powered by Huawei Ascend NPUs.
Q69	ModelArts	[Formulas & Constants] Huawei Cloud end-to-end AI development platform supporting automated model generation and workflow scheduling.
Q70	ONNX	[Formulas & Constants] Open Neural Network Exchange (.onnx); enables framework interoperability and cross-platform inference acceleration.

Page Answers: *Master Answer Key & Deep-Dive Technical Explanations*